

assign an IP address to the machine upon connecting to the network. Clicking on the Release button will release the IP address, and the Renew button requests a new IP address from the DHCP server.

Primary and Secondary WINS Server

These settings specify the IP address of the Primary and Secondary WINS Server, if present. The WINS servers translate the NetBIOS names into the corresponding IP addresses.

Lease Obtained and Leased Expired

These values show when the current IP address was obtained and when it is due to expire. You can use the Release and Renew buttons to release and renew your current IP address. In any case, the DHCP client automatically attempts to renew the lease when 50 per cent of the lease time has expired.

Ping

The Ping utility will let you determine whether a specific IP address is accessible. It works by sending a data packet to the specified address and waiting for the reply. Ping is a simple, yet highly configurable tool.

Trace Route

You can troubleshoot your Internet connection with the Trace Route option. To use Trace Route, type *tracert* in the command line. Tracert traces every hop (router on the Inter-



Use *tracert* to troubleshoot your routing problems

net) that a packet takes to reach its destination, giving the path and the amount of time taken between each hop. On a LAN, this can be used to diagnose problems in routing. Use it on the Internet to choose the nearest and fastest download sites.

Netstat

Use Netstat to display protocol based statistics, as well as all current inbound and outbound connections. Netstat can be configured to display



Netstat shows you inbound and outbound connections through all the ports

all connections and listening ports and Ethernet statistics, as well as statistics on a protocol basis. This would include statistics for TCP, UDP, ICMP and IP. This is a very useful command line utility that lets you know who is accessing the machine.

IP Config

IP Config is a very useful tool that displays the entire TCP/IP

configuration in your machine. The IP Config utility displays all the information in the Win IP Config utility, but on the command line. To obtain all the information, use the *ipconfig/all* command.

Net

The Net utility in the command line contains a host of other interlinked utilities. You can choose to do basic network configuration, diagnostics, inspect print queues, as well as connecting and disconnecting to the network from the command line. Type in *net /?* on the command line to access more information on the different commands within the Net utility.

For command line utilities, in order to be able to read the text output one screen at a time, type in the command through the More paging command—this would be of the form *net /? | more*.

LAN protocol

The exchange of information takes place through established protocols. Windows 98 supports the following transport protocols:

TCP/IP: TCP/IP is installed by default in Windows 98. If your computer is located on an interconnected network with diverse hardware and operating systems, you would need to use the TCP/IP protocol. TCP/IP is the standard protocol for the Internet, and also the de-facto standard for LAN communications.

NWLink IPX/SPX Compatible Transport: This is the standard network protocol

that is used to communicate with NetWare networks. It supports routing too. IPX/SPX is needed only if your computer is connected to, or communicates with a NetWare network.

NetBEUI: You need to install this protocol only if your network uses NetBEUI as a transport protocol.

Generally, for a small network, all you need to have installed is the TCP/IP protocol suite that is installed by default in Windows 98.

Driving for DOS

If you use DOS applications, you may not be able to use files and shared resources that are located across the network, since these programs need a full path to the files, starting with a drive letter. Within Windows, you can map folders that are shared across the network, to free drive letters to your computer. Use Explorer to get to the shared folder and then right-click and choose 'Map Network Drive'. Select any free drive letter in the configuration screen. Now you can



Mapping frequently used network shares